

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Summer 2024

Course: B. Tech.	Branch: CIVIL	Semester: V
Subject Code & Name:	Advance Environmental Engineering	BTCVPE506A
Max Marks: 60	Date: 12/07/2024	Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

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|---|------------|-----------|
| A) Design a septic tank for 75 users. Assume necessary data. | CO I/BT V | 06 |
| B) Rewrite the equation that is used for modelling of self-purification of stream process. Explain Each and Every term used in it. | CO I/BT VI | 06 |
| C) Classify Low-cost treatment methods. Explain Bacterial Algal Symbiosis in detail with the help of neat sketch. | CO I/BT IV | 06 |

Q.2 Solve Any Two of the following.

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|---|--------------|-----------|
| A) Explain effect of industrial waste discharge on STPs. | CO II/BT IV | 06 |
| B) Explain processes of neutralization of Alkaline industrial wastewater. Discuss any two methods. | CO-II/BT IV | 06 |
| C) Explain General Characteristics of Industrial Wastewater. | CO II/ BT IV | 06 |

Q. 3 Solve Any Two of the following.

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|--|---------------|-----------|
| A) Consider following data:
A industry wants to recycle the industrial wastewater containing heavy metals like Hg and Pb. Industry has enough funding to choose best suited technology as a tertiary treatment. Industry authorities wish to have recycled water quality to be amongst best.
Propose a suitable heavy metal removal technology for secondary treated waste of above industry. | CO-III/ BT VI | 06 |
| B) Compare Nitrification and Denitrification of wastewater | CO-III/ BT V | 06 |
| C) Explain how Ion Exchange can be used for tertiary treatment of Industrial Wastewater. | CO-III/BT IV | 06 |

Q.4 Solve Any Two of the following.

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|---|--------------|-----------|
| A) Categorize vector Transmitted diseases. Explain any one in detail. | CO-III/ BT V | 06 |
| B) Explain Life cycle of house fly. Also examine fly control | CO-III/BT IV | 06 |

w.r.t. different stages of life cycle

- C) **Explain** chain of disease transmission CO -III/BT IV 06

Q. 5 Solve Any Two of the following.

- A) **Examine** suitability of septic tanks in environmental sanitation with help of neat sketch. CO-IV/BT VI 06
- B) **Explain** water quality criteria prescribed by Central Pollution Control Board (CPCB) CO-IV/BT IV 06
- C) **Evaluate** suitability of Eco-San toilet. Also draw a neat sketch CO-IV/BT VI 06

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